

Predict-AI: Resilient Supply Chains

via Motor Health Monitoring

Prototype Submission For:

[Smart Supply Chains] Resilient Logistics & Dynamic Supply Chain Optimization

1. The Challenge

- Unplanned electrical machine and motor failures instantly disrupt the supply chain.
- Manufacturing delays and halted production lines lead to severe bottlenecks in logistics.
- Reactive maintenance causes unpredictable downtime, missed quotas, and increased costs.
- Lack of real-time machine health monitoring leaves logistics networks vulnerable to sudden shocks.

2. Our Solution

- Machine Learning-based Predictive Maintenance AI System.
- Continuously collects and analyzes live Modbus data from VFDs (frequency, current, temp).
- Predicts mechanical and electrical faults before they escalate to catastrophic failures.
- Shifts the paradigm from reactive to proactive maintenance, ensuring continuous operation.

3. System Architecture

- Data Collection Layer: Modbus RTU protocol communicating with industrial VFDs.
- Data Processing Layer: Python-based data cleaning, normalization, and feature extraction.
- ML Inference: Custom AI predictor analyzing temporal patterns to detect anomalies.
- Interface: Real-time dashboard for visualization, control, and early-warning alerts.

4. Prototype Demo & Capabilities

- Live data scanning and automated VFD register mapping.
- Proactive, automated alerts for abnormal machine behaviors predicting imminent downtime.
- Robust data logging architecture designed to continuously store and improve AI training.
- Minimal invasive setup that allows easy plug-in integration with existing industrial setups.

5. Business Impact for Supply Chains

- Dramatically reduces unplanned downtime, ensuring smooth production timelines.
- Enables predictable maintenance scheduling without needlessly halting active logistics pipelines.
- Prevents missed delivery targets and preserves supply chain reliability.
- Lowers long-term operational costs and builds a resilient, shock-proof production environment.

6. Future Scope

- Cloud integration for centralized, multi-factory predictive monitoring.
- Automated API triggers directly linking machine health to supply chain management platforms.
- Auto-ordering of replacement parts triggered the moment a machine's AI health score drops.
- Integration with edge-computing devices for ultra-low latency fault detection.